

Reconstructing 3d Segmentation of the Urban Environment using Street View Images

ストリートビュー画像を用いた都市環境の3D再構築とセグメンテーション分析法

The 34th Annual Meeting of The Geographic Information System Association Conference

Toyama, Japan

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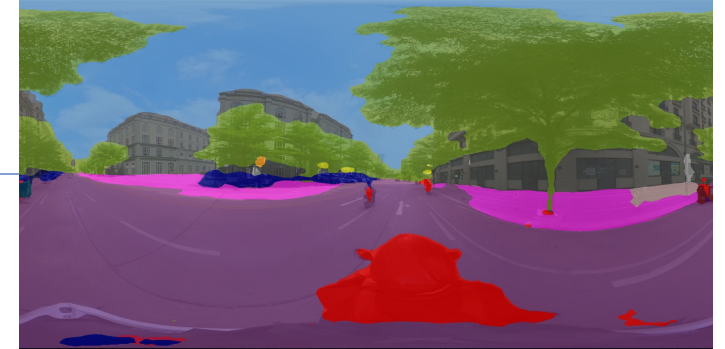
Background – Panoramic images in urban studies

- Street view images are widely used for urban researches



Street View images

Quantifying urban scene

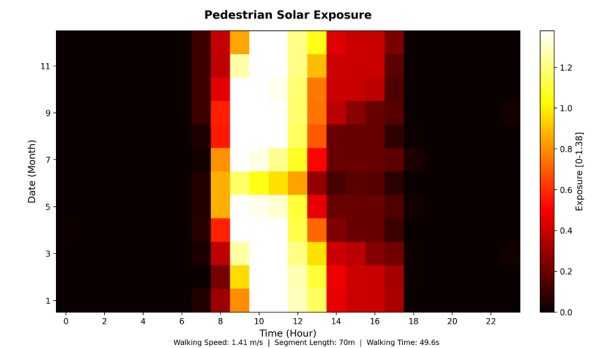


Visual perception study

Safe
Attractive
Beautiful
Green



Urban climate assessment



Solar exposure assessment

Potential urban assessment using 360 images

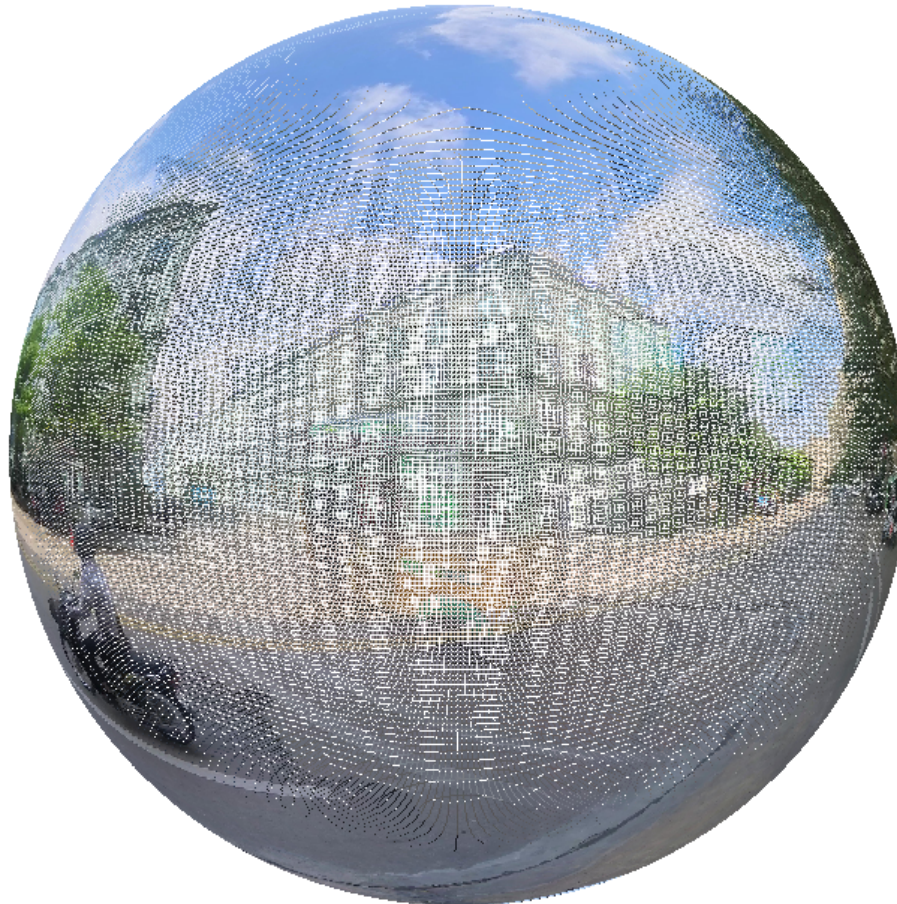
Background – Panoramic images properties

- Street view images have its own unique properties but also limitations

Panoramic images

Benefits

- Can be easily obtained
- Affordable and available from multiple sources
- Able to capture the whole surrounding environments

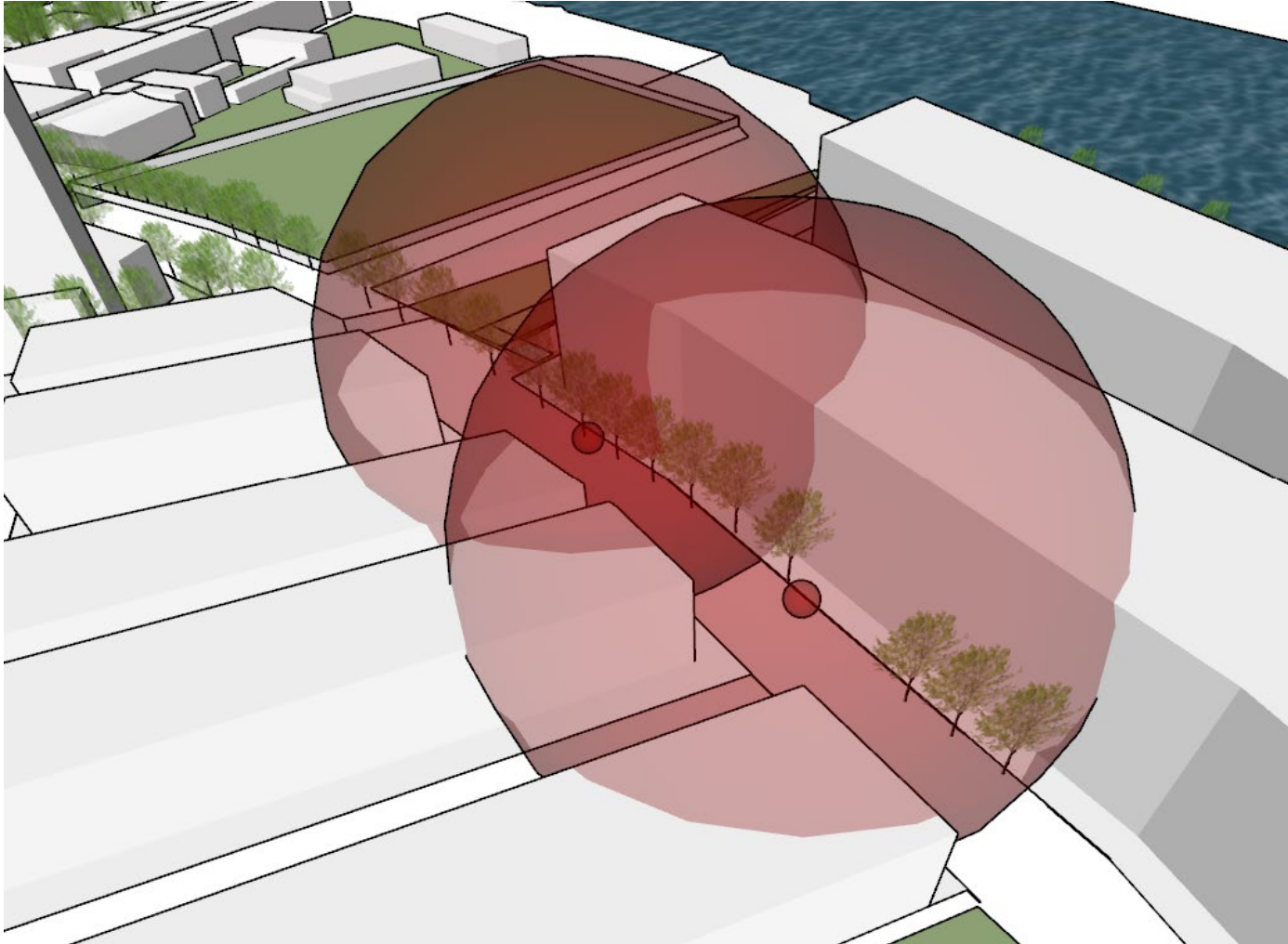


Drawbacks

- Lacks the depth data of the environment
- Illustrates the car's perspective not the pedestrian's perspective

Background – Utilizing data from sequential images

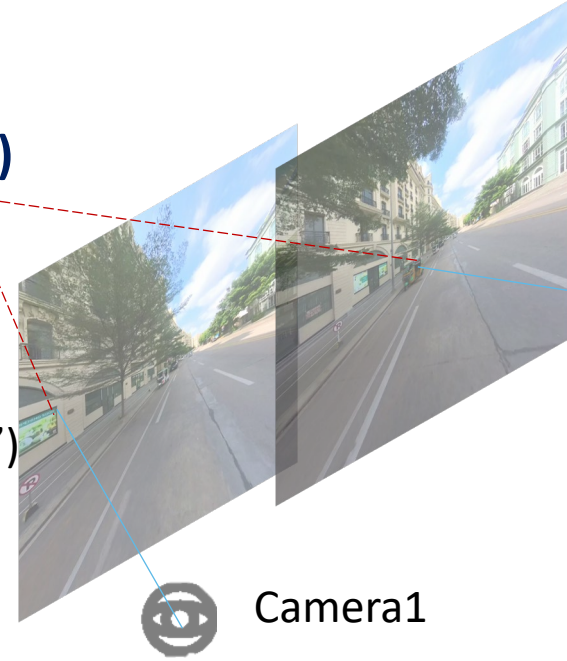
- When combine SVIs together, new data (depth) can be obtained



Structure from motion principle

$A(x, y, z)$

$U'(x', y')$



Camera2

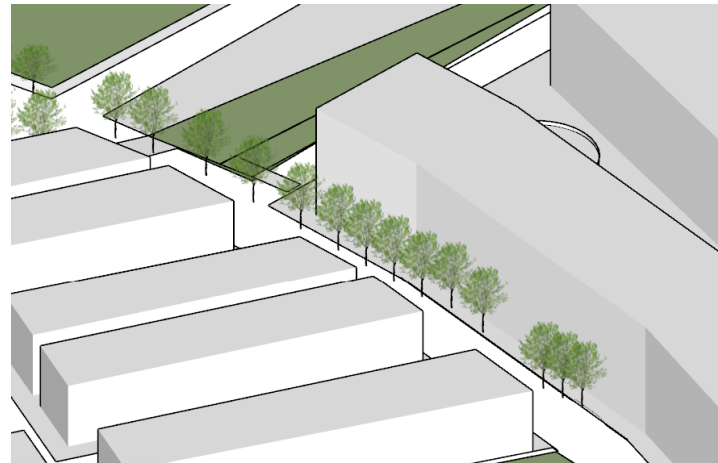
Camera1

Tomasi-Kanade (1992)

Tomasi, C., & Kanade, T. (1992). Shape and motion from image streams under orthography: a factorization method. International Journal of Computer Vision, 9(2), 137-154.

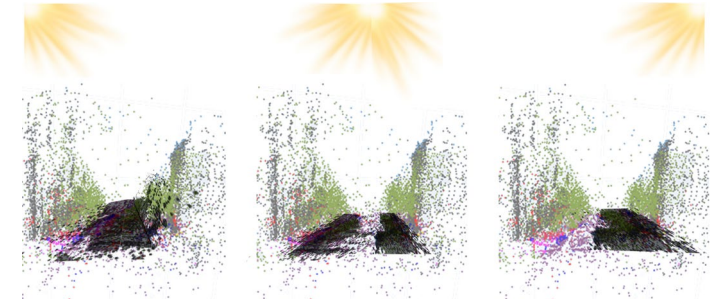
Background – Understanding 3d reconstruction

- Application of 3d reconstruction



Reconstructed 3d

Simulations

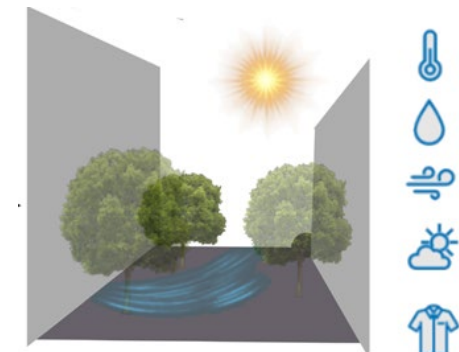


Spatial perception

Safe
Attractive
Beautiful
Green



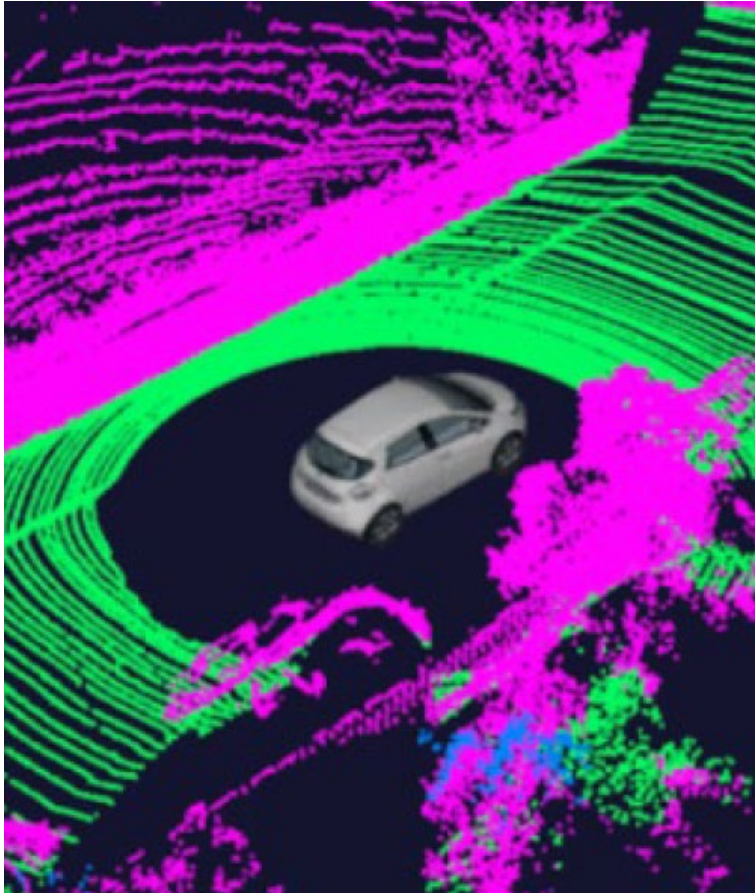
Thermal comfort assessment



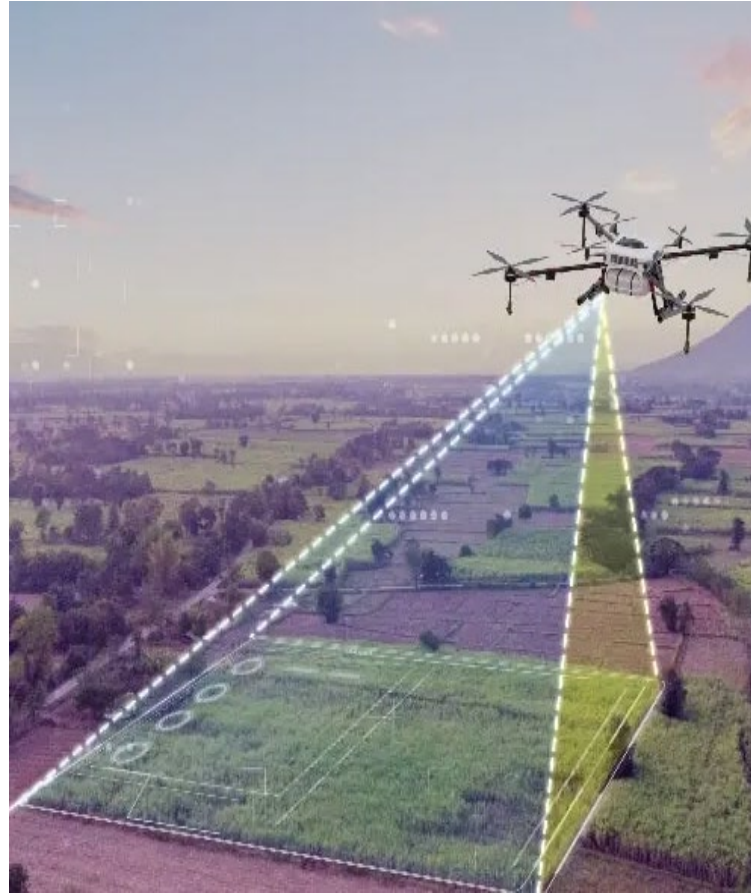
Potential urban assessment using 3d data

Background – Current 3d reconstruction method

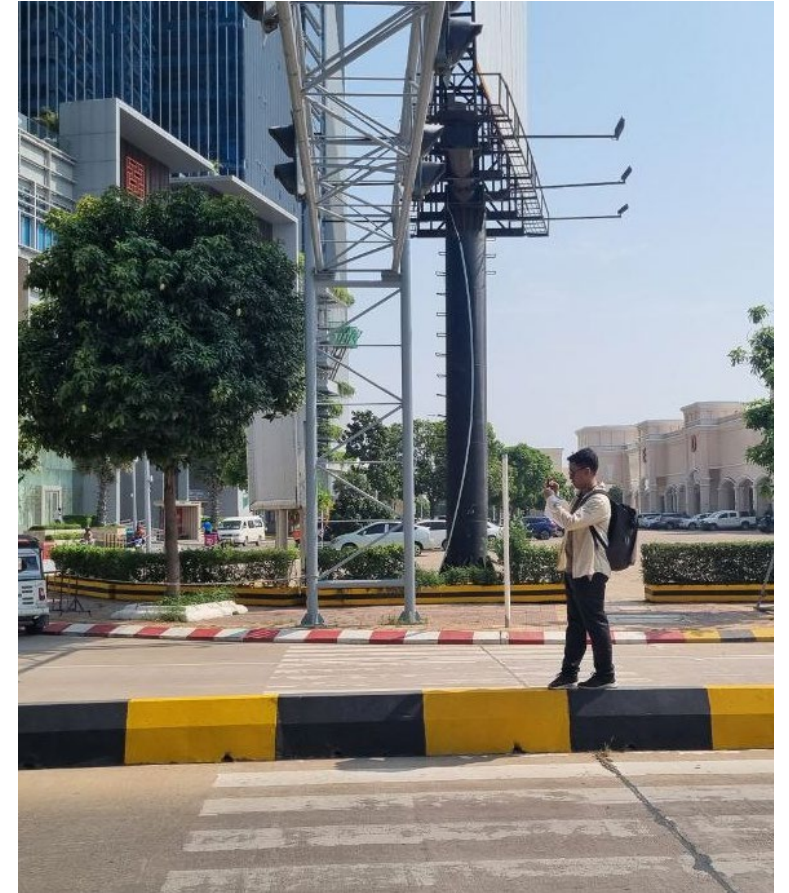
- The process require time and investments, cannot keep up with urban growth speed



Lidar Scanning



Drone Photogrammetry



On-site Survey

Background – Digital Twins and reality gap

- The result lacks fine-grain details and actual human experience of the city



Digital Urban Twins (n.d.) [Flanders region, Belgium, DTC of Flanders region, Belgium] [Online image].

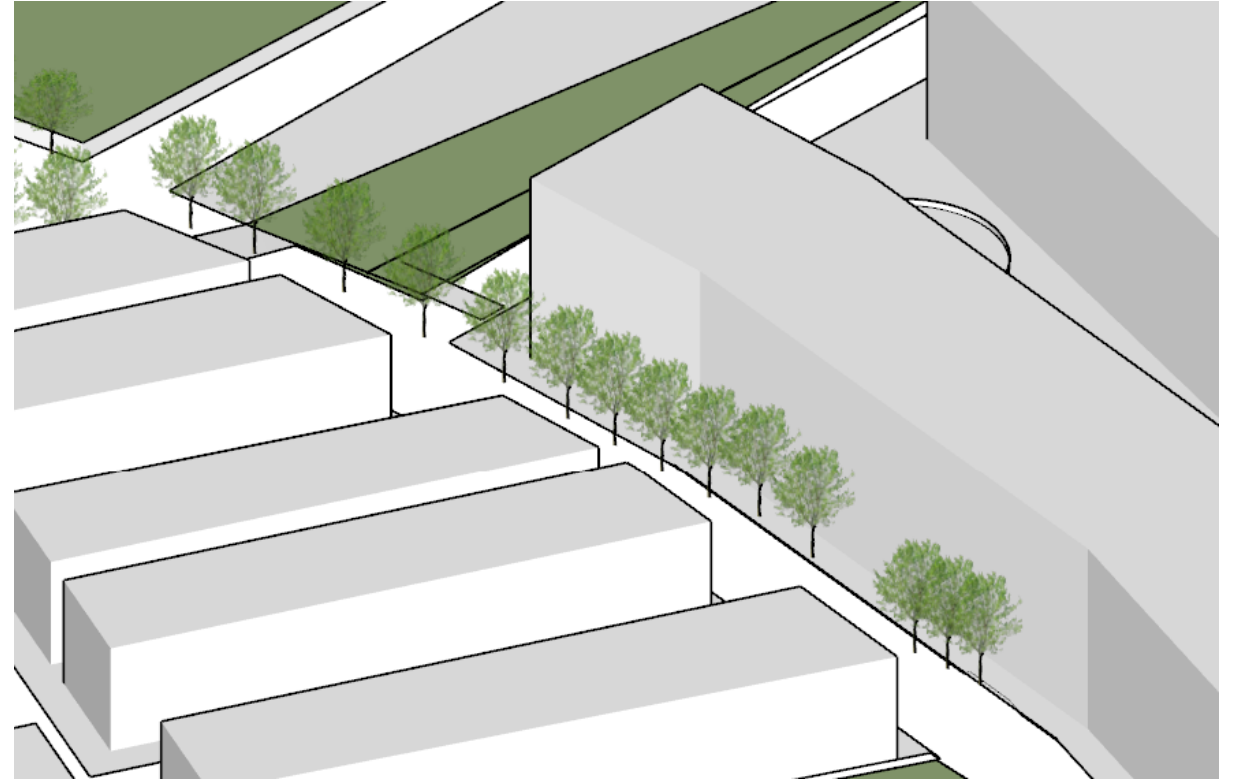
Available at: <https://www.digitalurbantwins.com/flanderstwin> (Accessed: 8 October 2025)

Google Earth (2024) Available at : (https://www.google.com/maps/@51.05404,3.72405,190m/data=!3m1!1e3?entry=ttu&g_ep=EgoyMDI1MTAwNi4wIXMDS0ASAFAw%3D%3D) (Accessed: 8 October 2025).

Reconstructing 3d Segmentation of the Urban Environment using Street View Images, Leng Monykandy, Gu Zongchao, Jun Cao / Southeast University / monykandyleng@gmail.com

Background – Expanding the image limitations

- Transforming images to 3d reconstruction



- Widely available
- Easy to obtain

Street View images

- Has a lot of data
- Accurate and applicable for diverse analysis

3d reconstruction

Objectives

Build a universal, quick research tool that's simple to use.

普遍的で、迅速、かつ使いやすい研究ツールを構築する。

- **Affordable**
低コスト : using only available 360 images
: 既存の360度画像のみを使用
- **Robust**
ロバスト性 : quick to produce assessment results
: 評価結果を迅速に生成
- **Applicable**
適用可能 : applicable in multiple aspects of urban research
: 都市研究の複数の側面に適用可能

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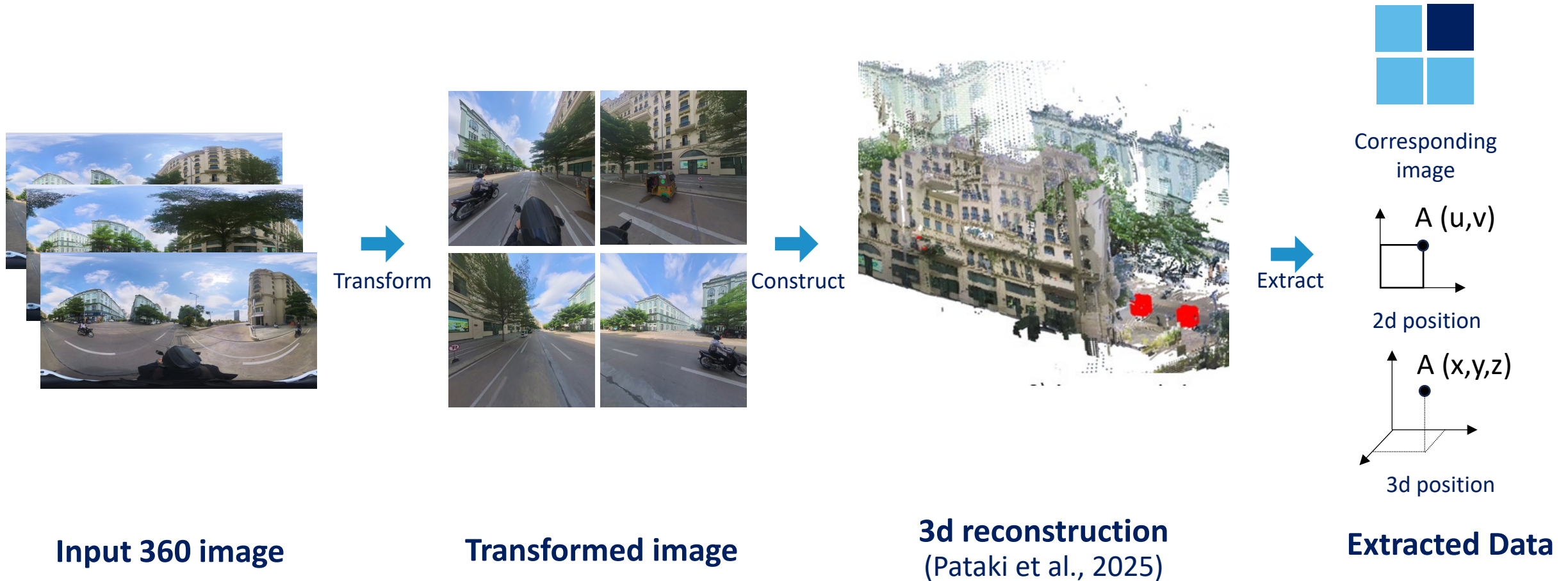
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Methodology – 3d segmentation

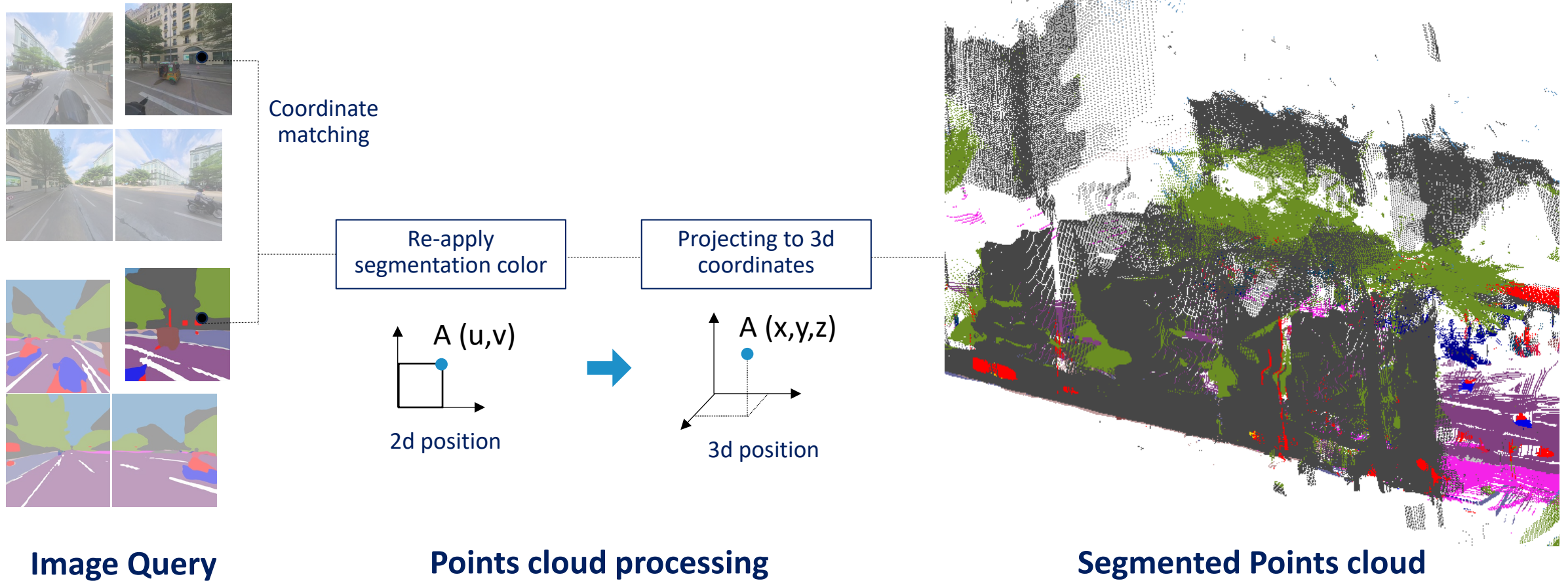
- 360 Images are processed to reconstruct points cloud segmentation



Pataki, Z., Sarlin, P. E., Schönberger, J. L., & Pollefeys, M. (2025). MP-SfM: Monocular Surface Priors for Robust Structure-from-Motion. In Proceedings of the Computer Vision and Pattern Recognition Conference, pp. 21891-21901.

Methodology – 3d segmentation

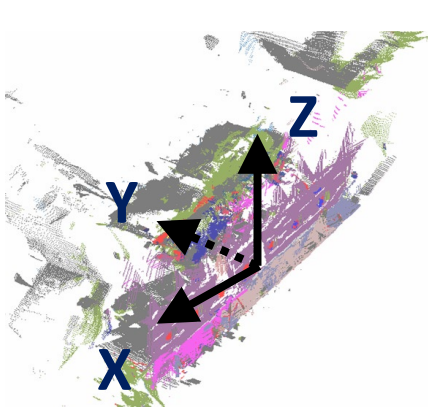
- 360 Images are processed to reconstruct points cloud segmentation



Pataki, Z., Sarlin, P. E., Schönberger, J. L., & Pollefeys, M. (2025). MP-SfM: Monocular Surface Priors for Robust Structure-from-Motion. In Proceedings of the Computer Vision and Pattern Recognition Conference, pp. 21891-21901.

Methodology – Reorient the points cloud

- Defining the world coordinates vector

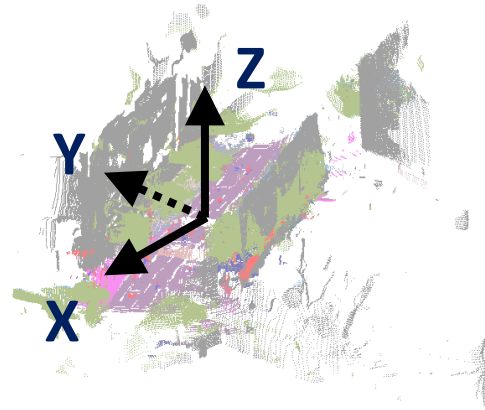


Output points



- Perpendicular to the ground
- Starting from the ground's centroid
- Parallel to the building's up vector

Assigning the up vector

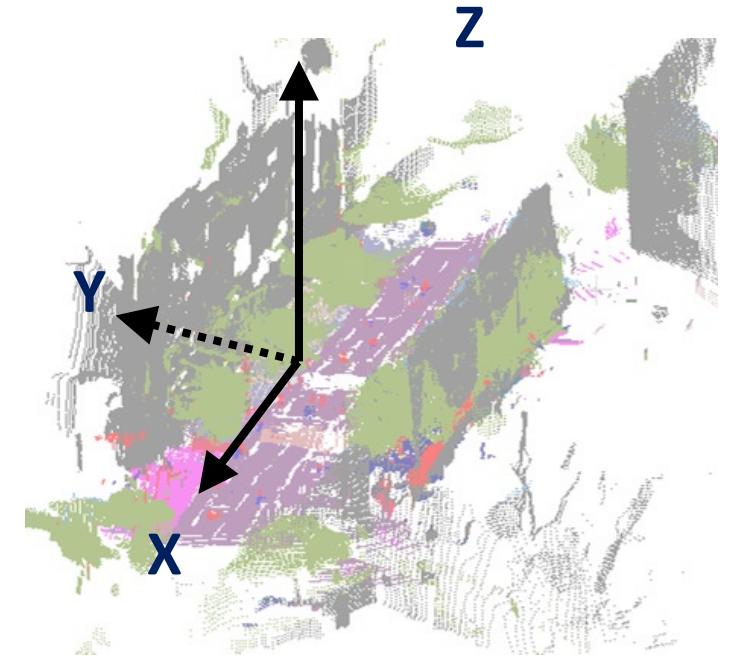


Z-axis oriented



- Detect the nearest buildings' facade
- Analyze the façade's dominant direction
- Parallel to the building's direction
- Perpendicular to the Z axis

Assigning the X axis

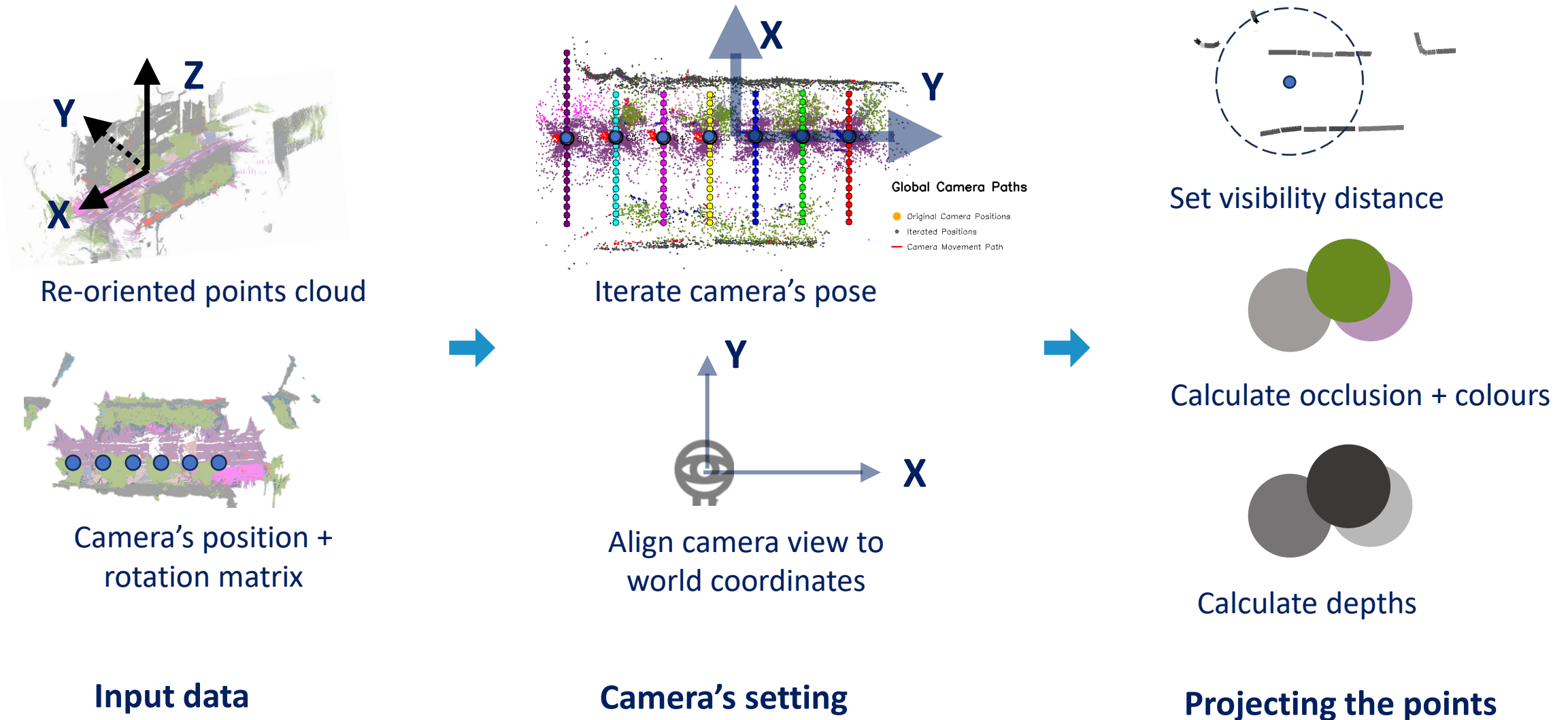


- Perpendicular to the X and Y axis

Assigning the last axis

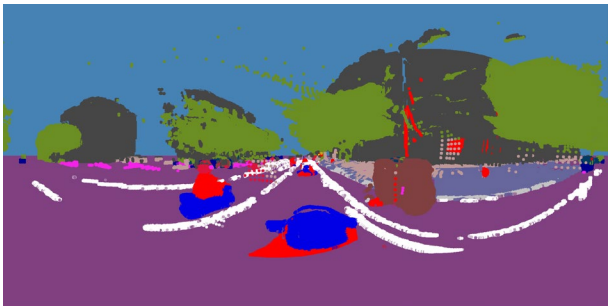
Methodology – Image / depth generation

- Camera's properties + points cloud data → new image generation

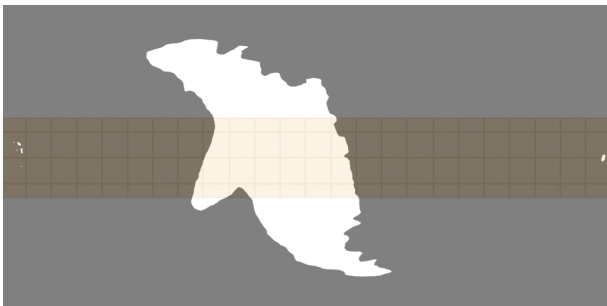


Methodology – Solar exposure assessment

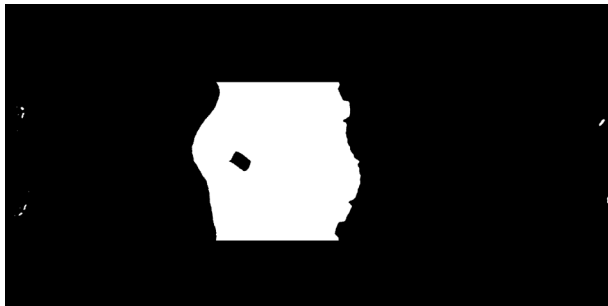
- Solar exposure assessment



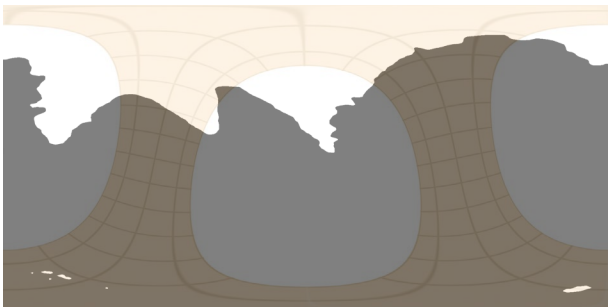
Segmented Image



Time calibrated sun's path



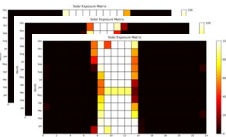
Corrected EQT (Equation of Time)



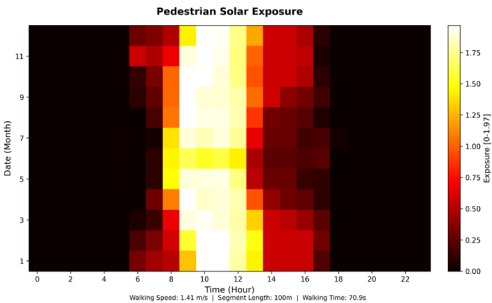
Altitude location



Visible sun area



Matrices summation



Solar exposure
(Zhu and Gu, 2022)

Pedestrian solar exposure



Zhu, H. and Gu, Z. (2022) A method of estimating the spatiotemporal distribution of reflected sunlight from glass curtain walls in high-rise business districts using street-view panoramas. Sustainable Cities and Society, 79, 103671.

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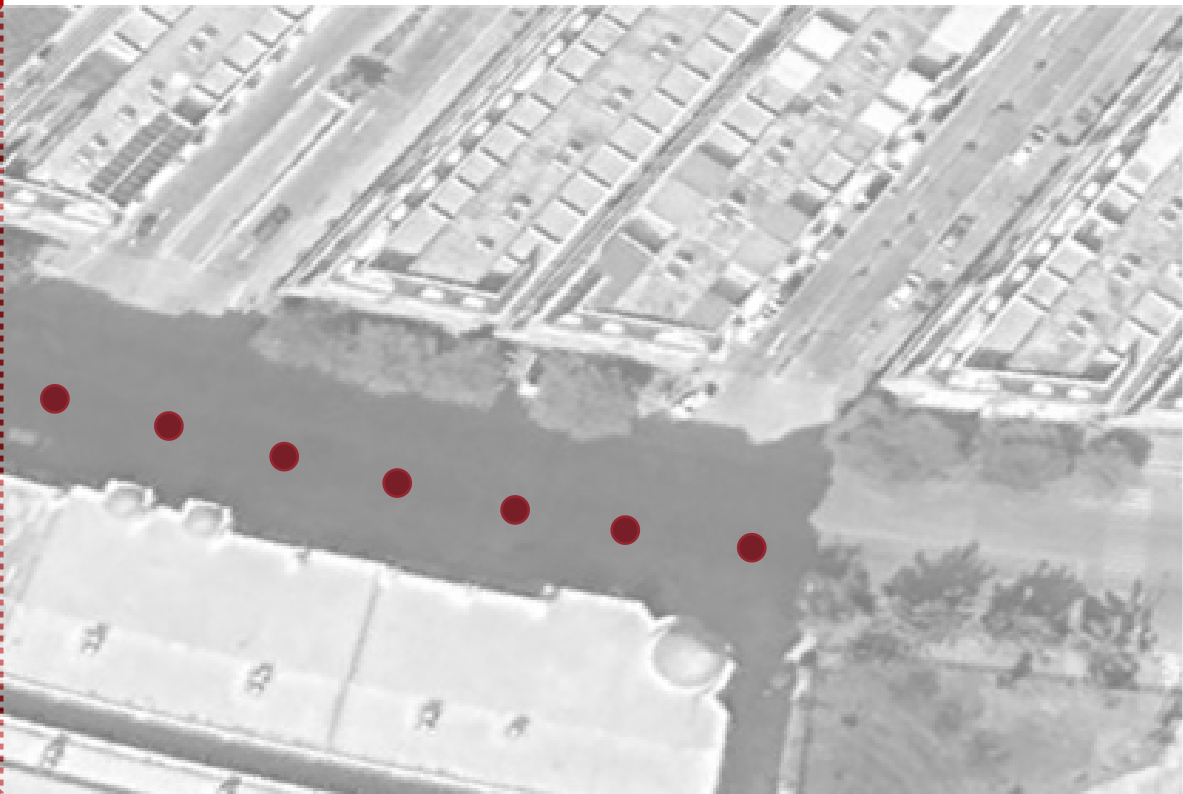
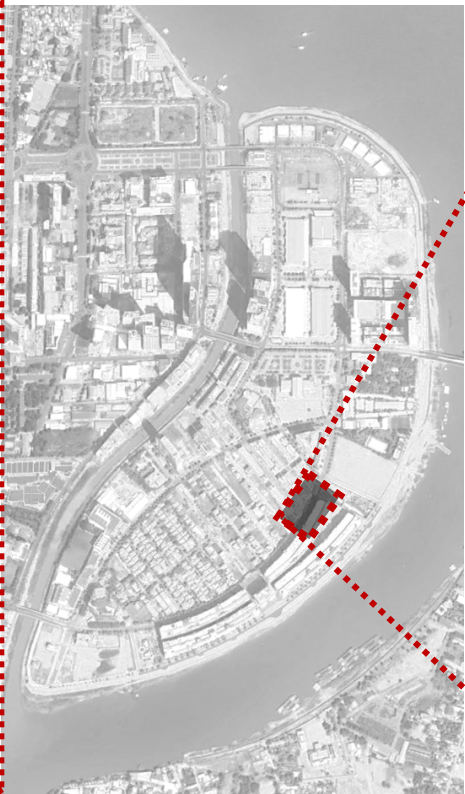
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Results – Study area



Parameters	Properties
Distance	10 meters
Camera	Insta360 X3
Resolution	5760 x 2880 pixels
Quality	96 dpi
Road Segment's Length	70 meters
Location	Koh Pich, Phnom Penh, Cambodia



Data parameters

Case study location

Camera's location

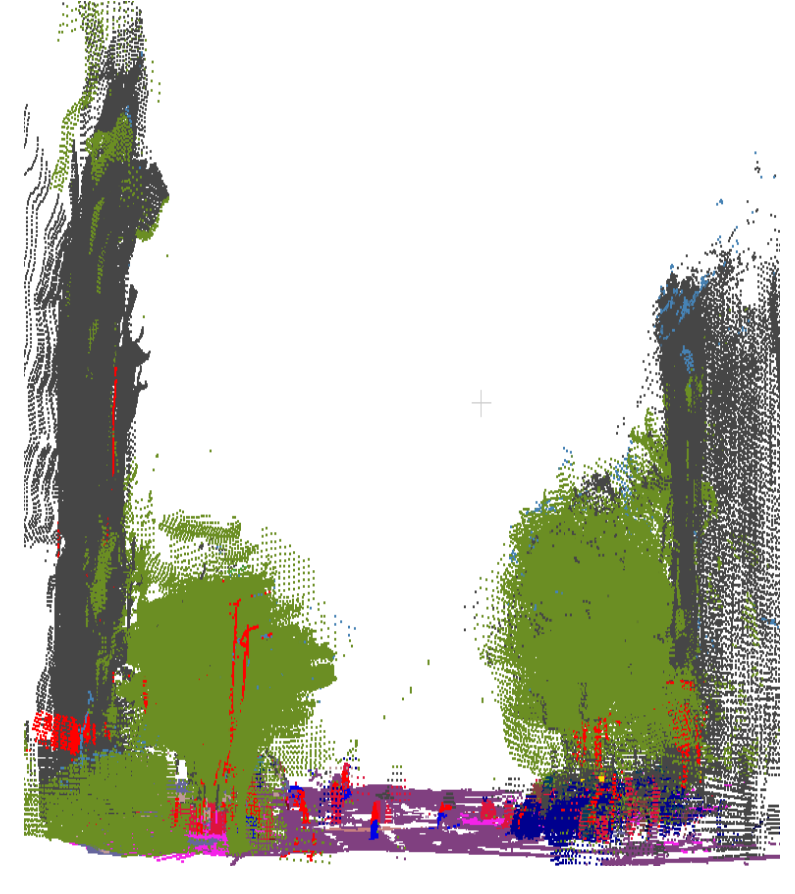
Results – Points cloud and reality



Real image



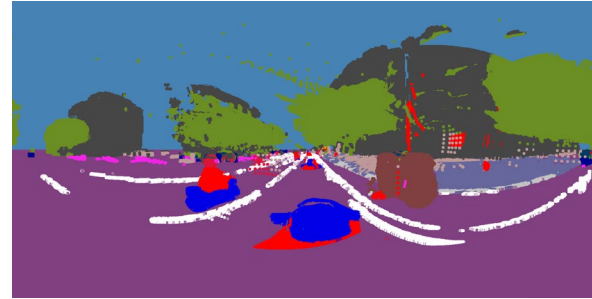
Points cloud image



Segmented points cloud

Results – Generated and captured image

Camera 1 (C1)



Camera 2 (C2)



Real Image

Segmented point cloud image

Projected point cloud depth image

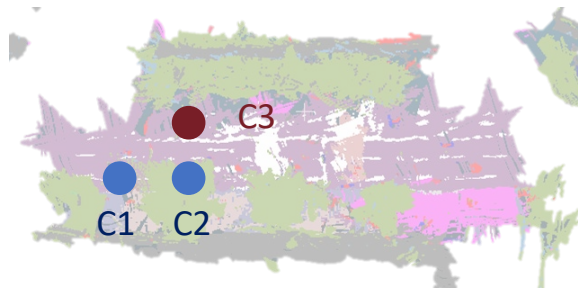
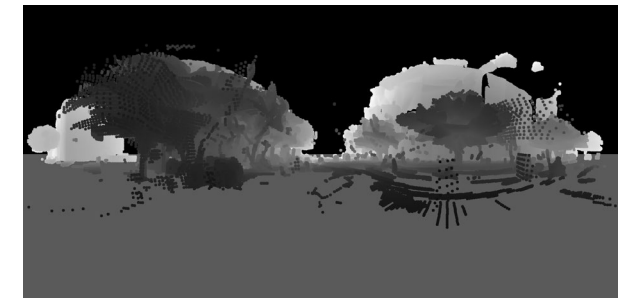


Image location

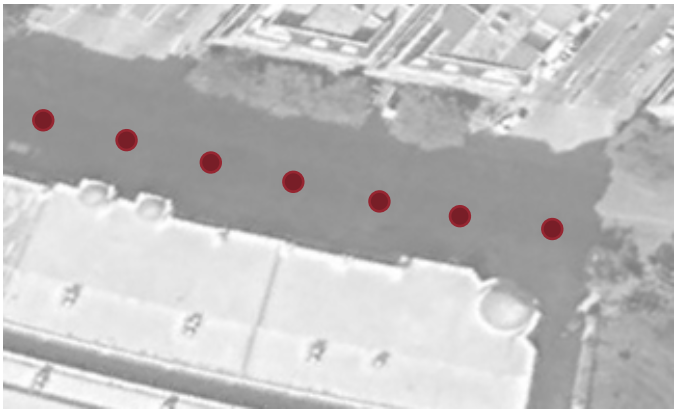
Generated image
(C3)



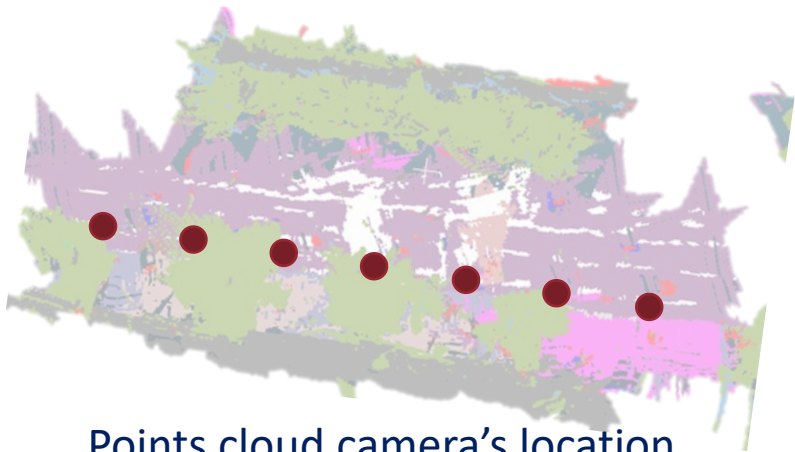
Segmented point cloud image

Projected point cloud depth image

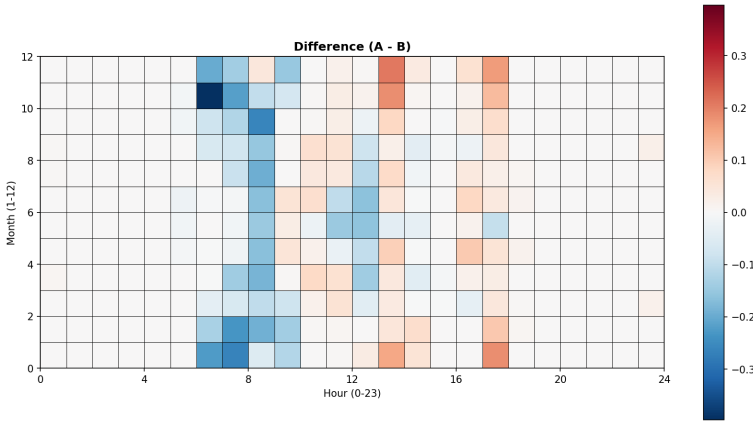
Results – Points cloud and 360 images (car's lane)



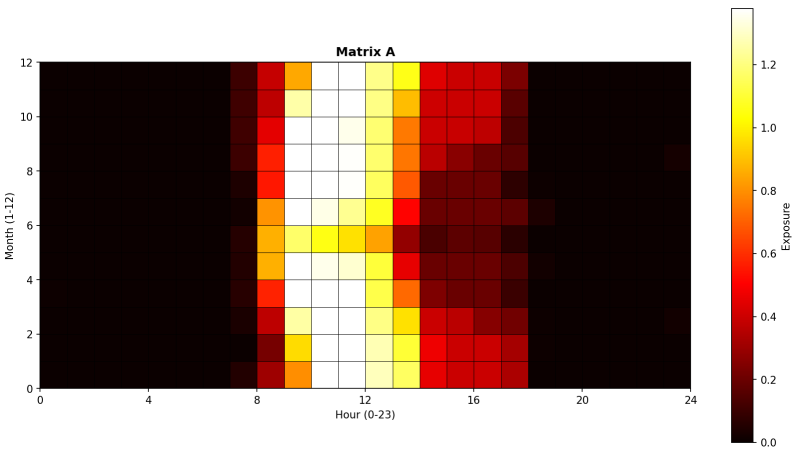
Original camera's location



Points cloud camera's location

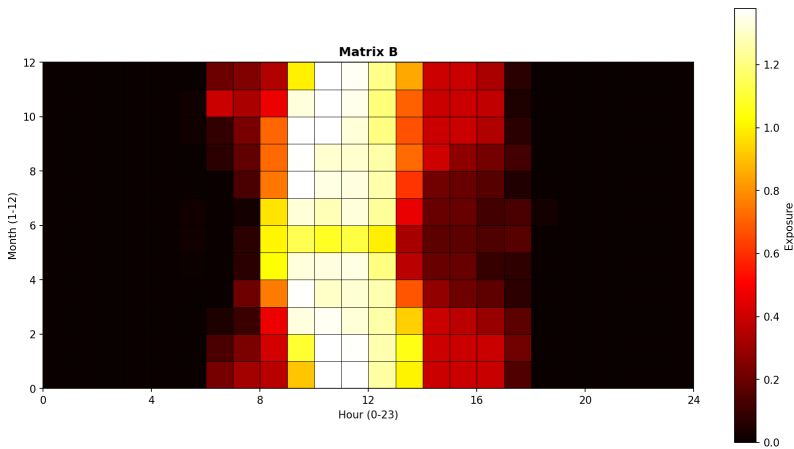


Signed difference



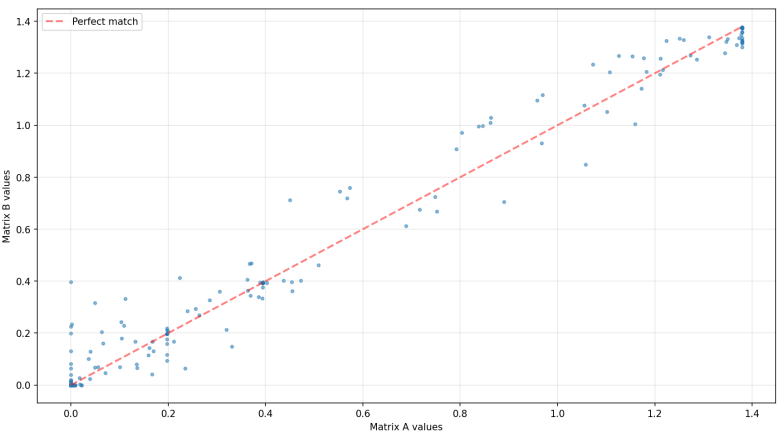
Pedestrian solar exposure

Results from original image



Pedestrian solar exposure

Results from points cloud

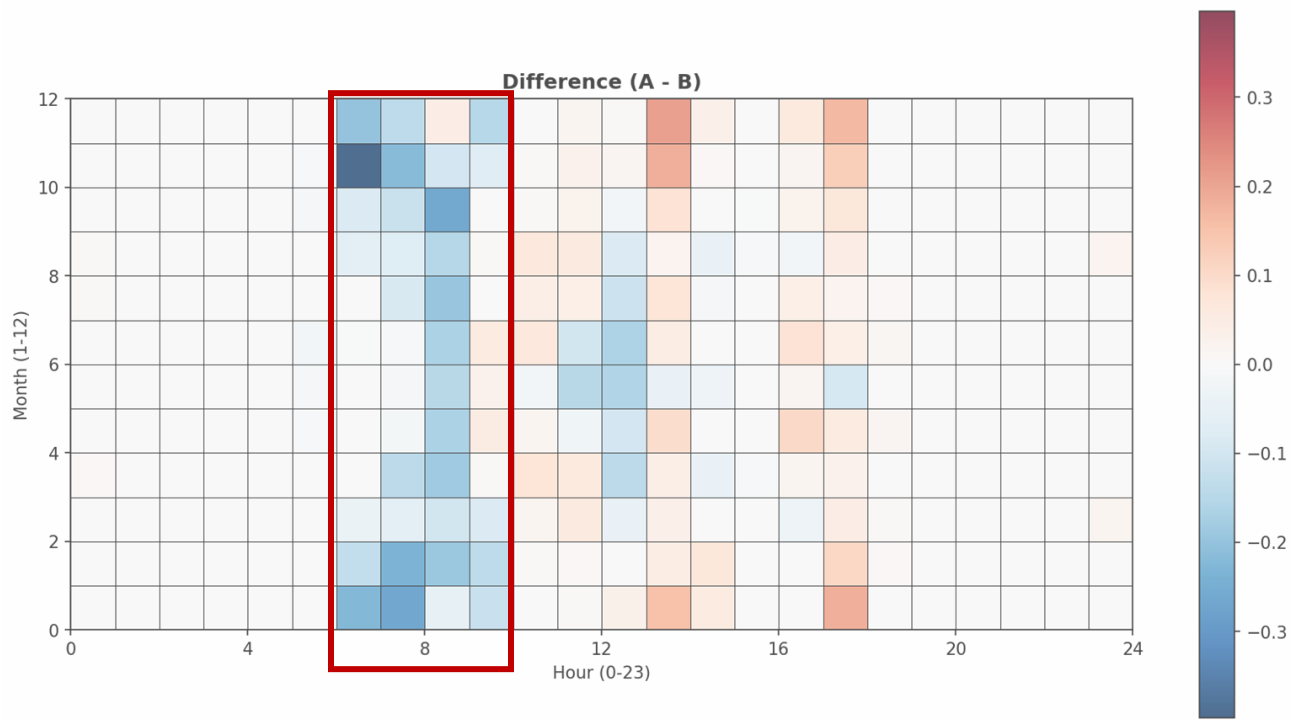


Overall Similarity: 85.11%

Matrix comparison

Results – Discussion

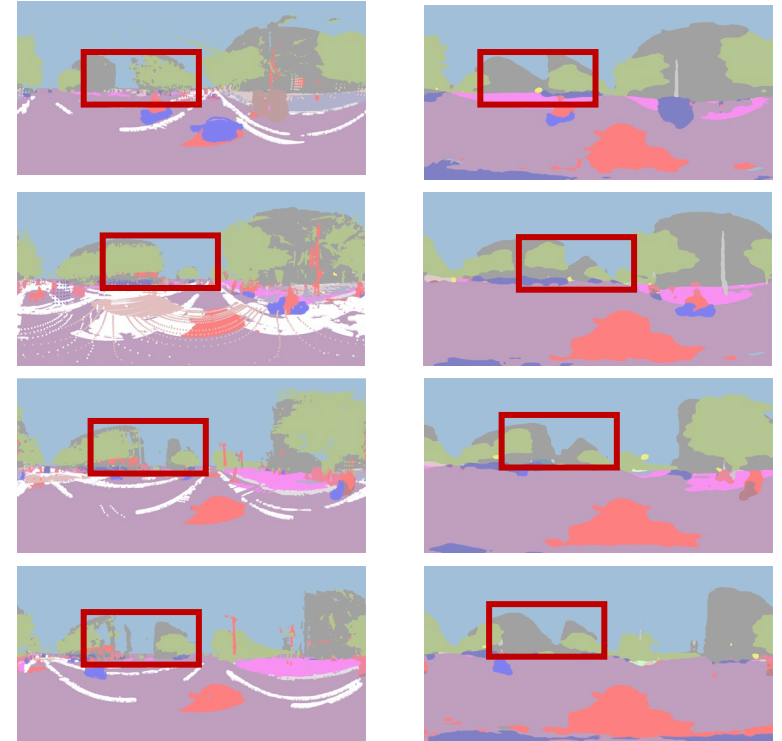
- Missing/inaccuracy in points cloud data



Max Difference : 0.3969
Min : 0
Max : 1.38
Overall Similarity : 85.11%

Points cloud data

Original data

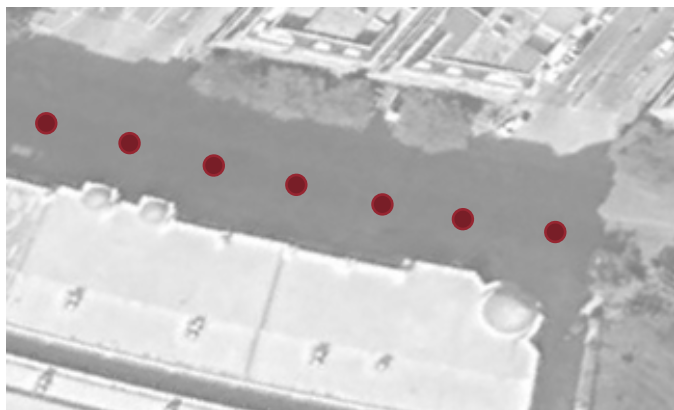


Reason of difference :

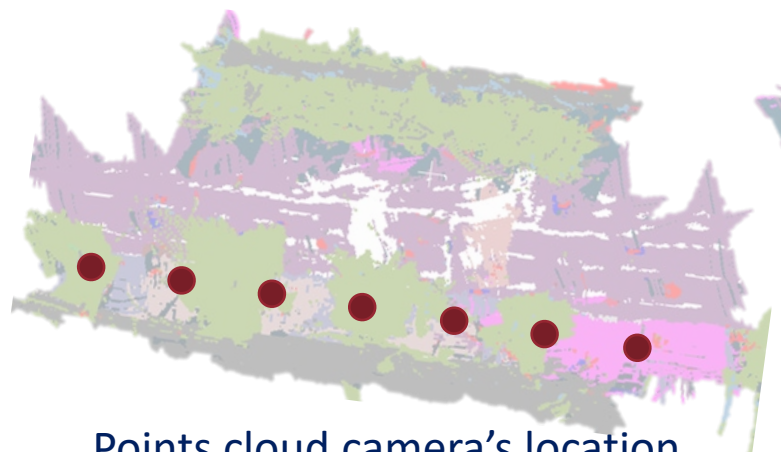
- Building's height inaccuracy
- Missing data during reconstruction
- Differences are spread out evenly

The solar exposure processed from the same location have the same results

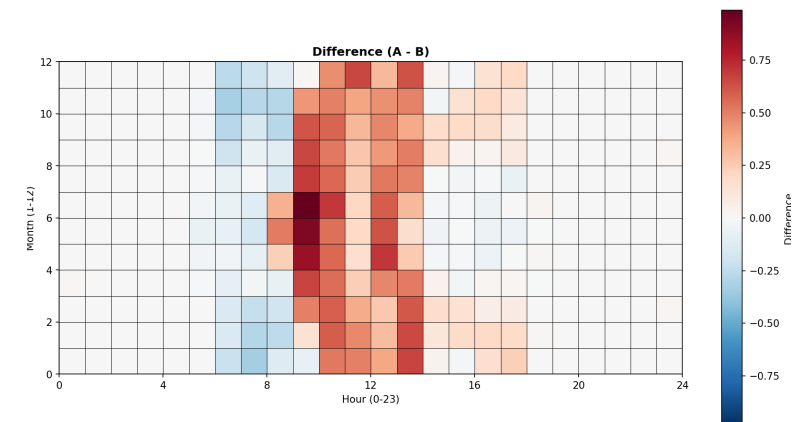
Results – Points cloud and 360 images (car & pedestrian lanes)



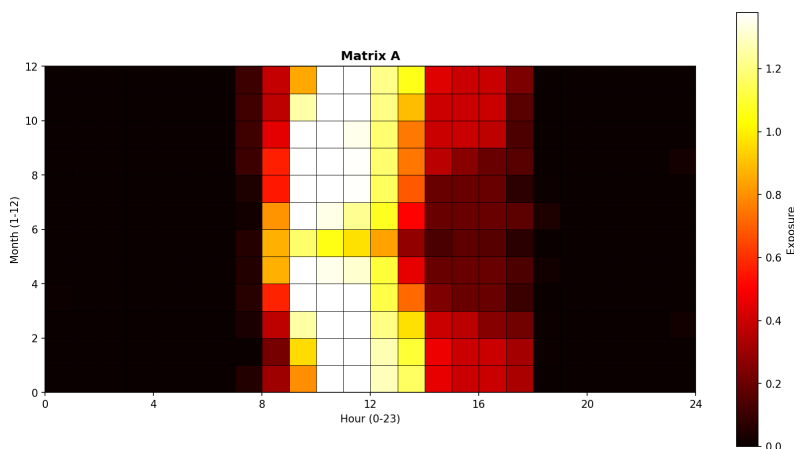
Original camera's location



Points cloud camera's location

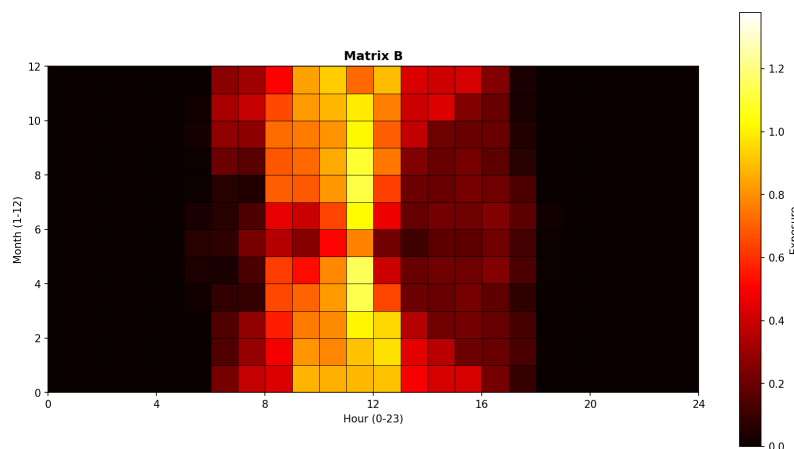


Signed difference



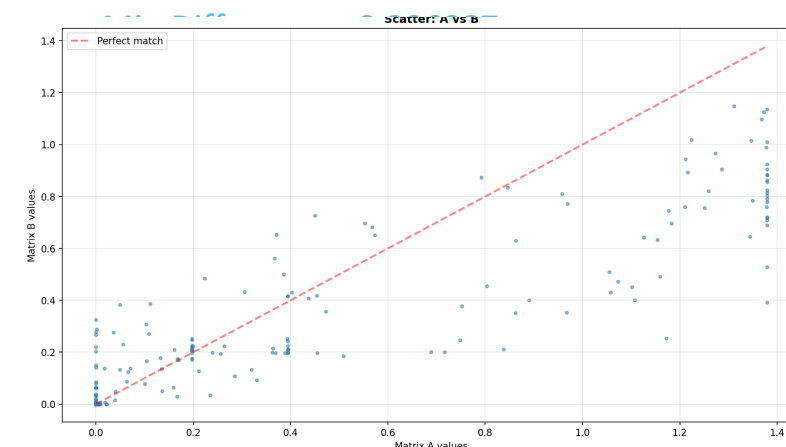
Pedestrian solar exposure

Results from original image



Pedestrian solar exposure

Results from points cloud

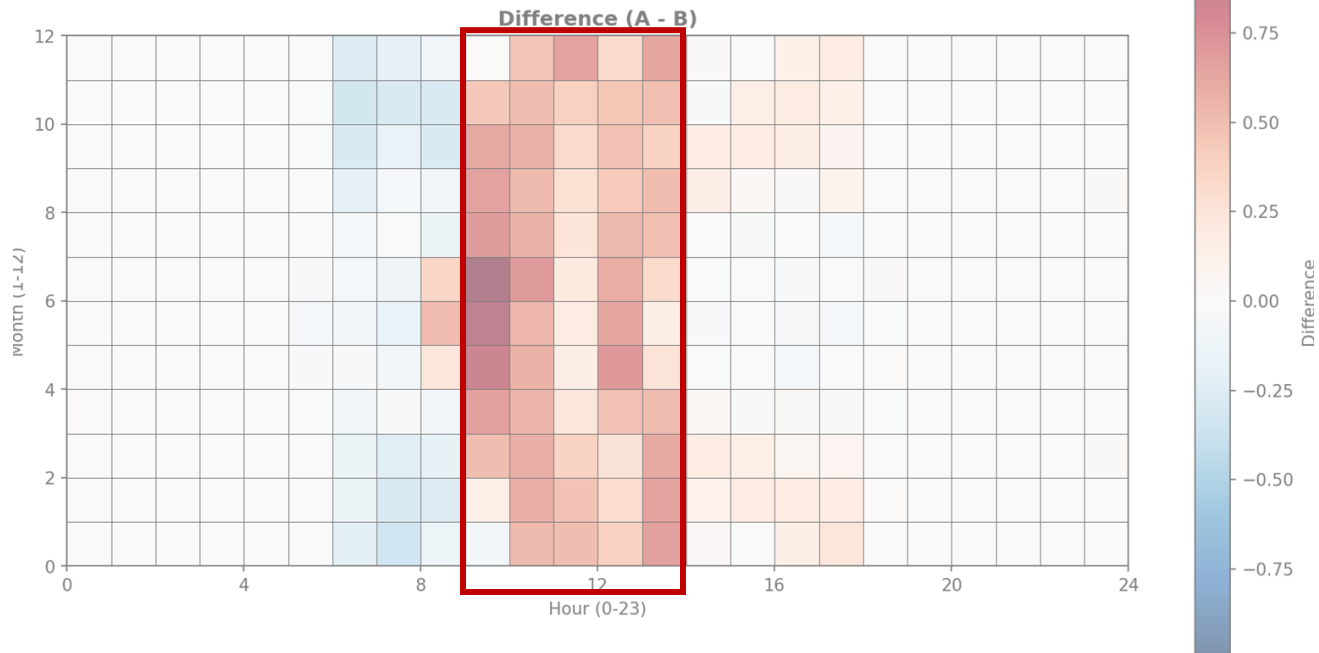


Overall Similarity: 75.63%

Matrix comparison

Results – Discussion

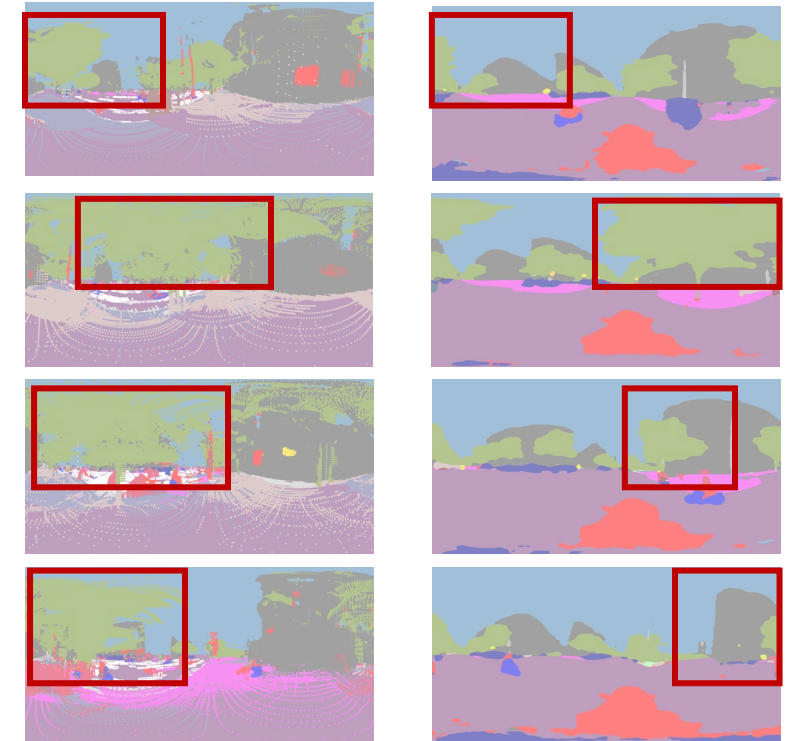
- Missing/inaccuracy in points cloud data



Max Difference : 0.9875
Min : 0
Max : 1.38
Overall Similarity : 75.63%

Points cloud data

Original data



Reason of difference :

- The images from car's lane doesn't have the shades of the trees
- The differences are concentrated in specific spot

The 360 images captured from the car's lane does not capture the actual pedestrian's data

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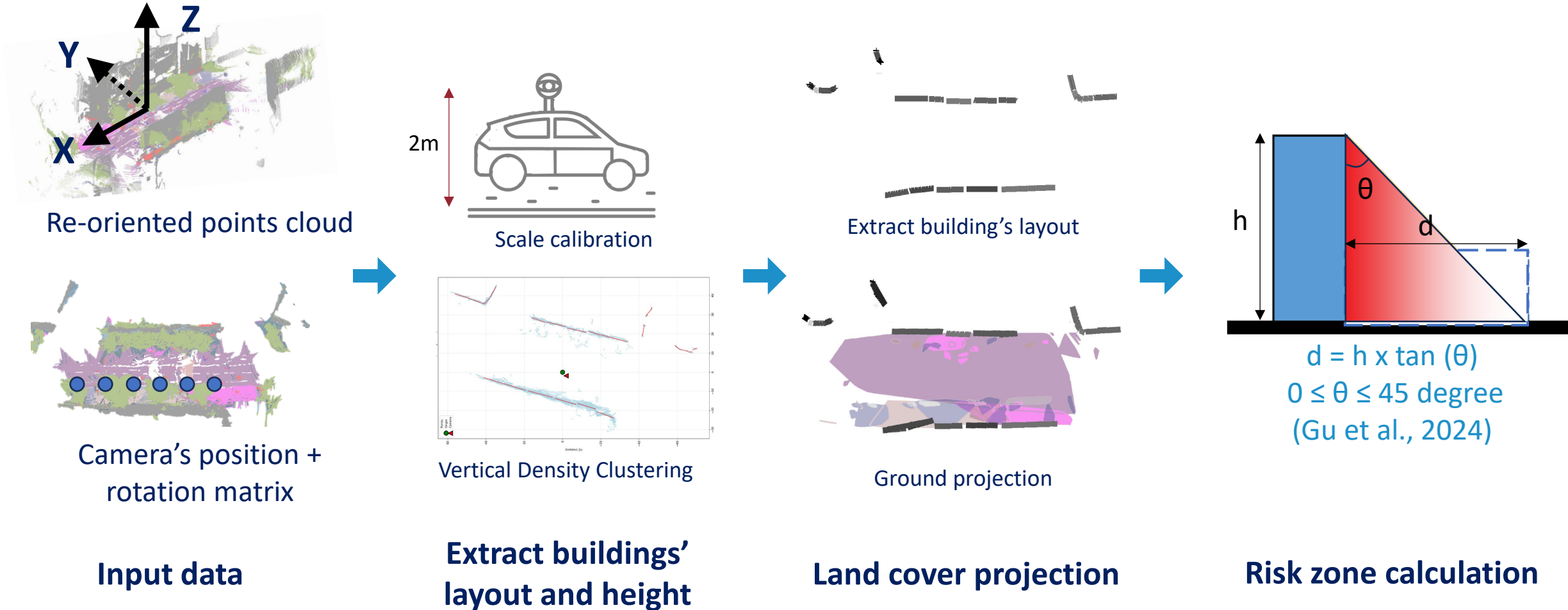
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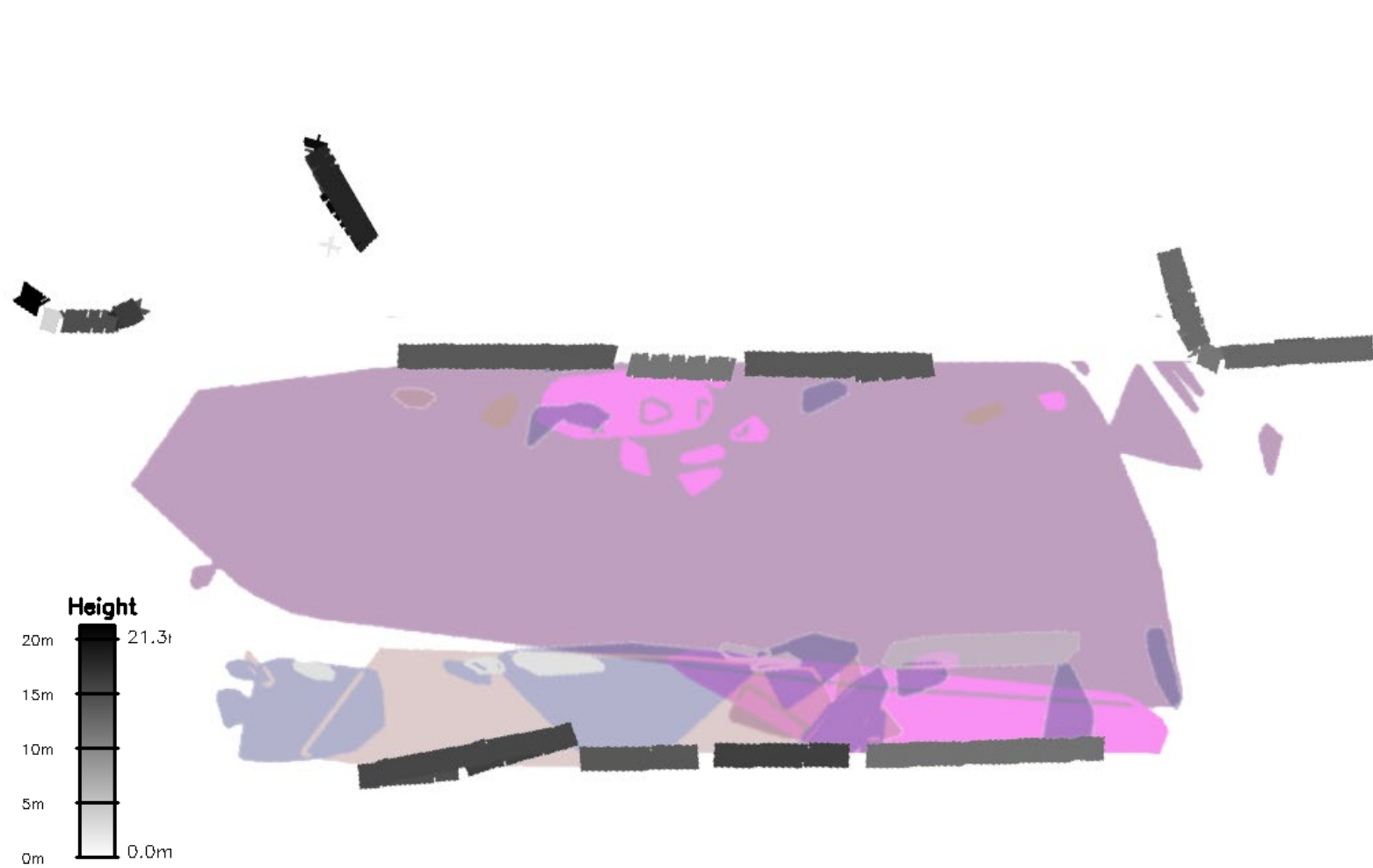
Methodology – Data processing

- Extract building's layout + Depth calibration → Seismic risk assessment

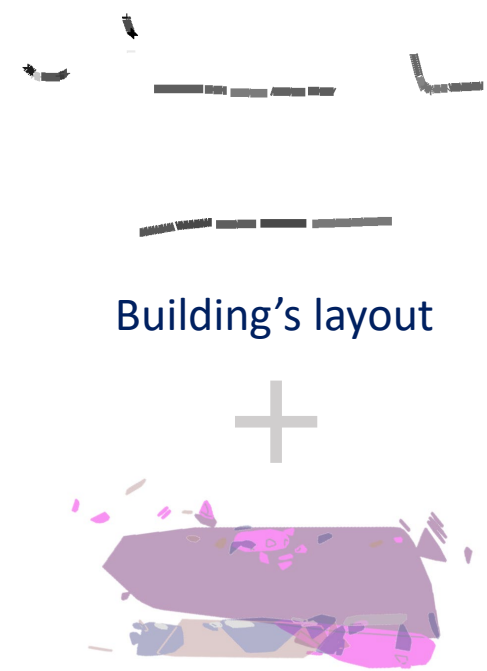


Gu, Z., Sun, X., & Zhu, H. (2024). A simple method of measuring the seismic risk of circulation space in urban residential streets based on street-view data. Paper presented at the Annual Meeting of the Geographic Information Systems Conference, October 2024.

Results – Projected landcover



Projected landcover with the buildings' layout

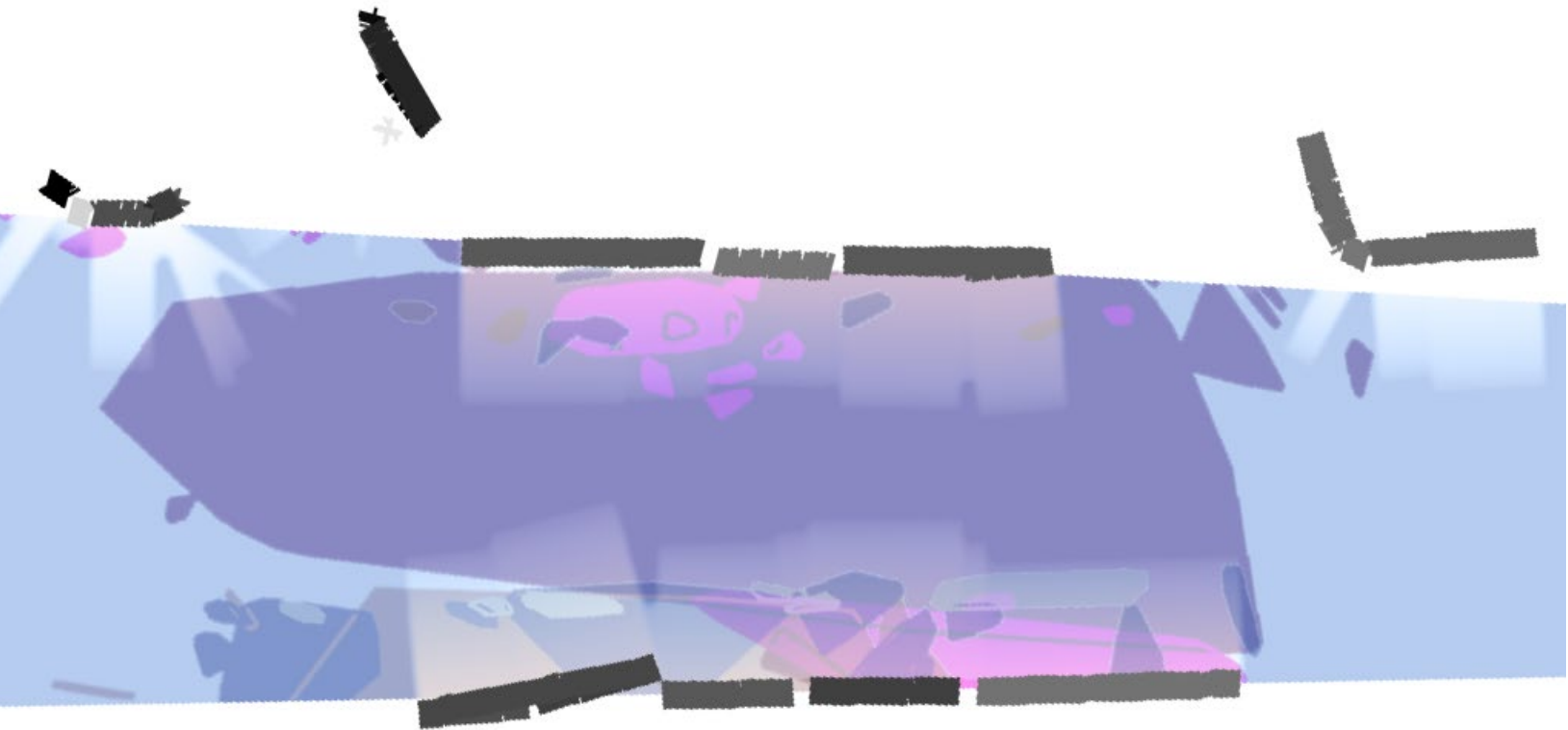


Building's layout

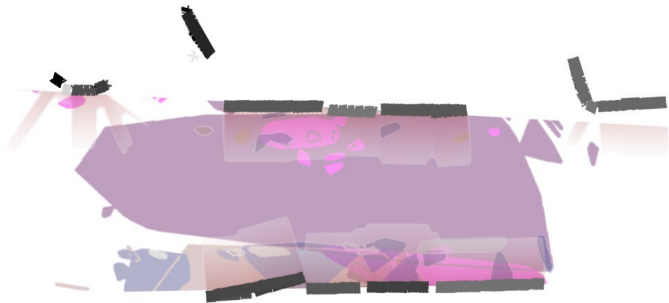
Ground projection

- | | |
|--|---|
|  Road |  Building |
|  Sidewalk |  Danger zone |
|  Fence |  Safe zone |
|  Barrier | |

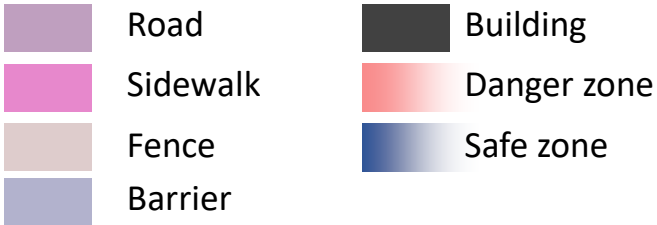
Results – Seismic risk assessment



Safe zone assessment



Risk zone assessment



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Summary – Contribution

- Segmented 3d points cloud from 360 images
- Illustrated that the solar exposure results of the car's lane is different from the pedestrian's path
- Estimate solar exposure with points cloud data
- Generate images using constructed points cloud
- Extracted building's layout from points cloud data

Summary – Limitation

- The reconstructed points cloud may result in incomplete data
- The 360 images property (able to capture the surrounding, is lost due to image transformation)
- The height of the building is not inconsistent because to transform the image to perspective, the buildings are cropped
- The sequential property of 360 images are not utilized
- The workflow assumes the ground is flat